CRT-591-MRXX

Card issuing Machine

Communication Protocol

V1.4

CREATOR (CHINA) TECH CO., LTD.

8/F, Block T3, Hualian City Business Center, No.1001 Nanshan Road, Nanshan Distr ict, Shenzhen,Guangdong,China.

TEL:+86 755 26710992 FAX:+86 755 26710105

Http://www.china-creator.com

Catalog

1. SERIAL PORT CONTROL SPECIFICATION: 5

1.1.1 *Communication format* 5
1.1.2 *Communication control method* 5
1.2 COMMUNICATION FORMAT AND CONTROL CHARACTER 5
1.2.1 *Send command package format* 5
1.2.2 *Successful responsive package format and character* 6
1.2.3 *Failed responsive package format and character* 6
1.3 ADDRESS SET FOR MULTI-UNITS COMMUNICATION: 7
1.4 COMMUNICATION METHOD: 8
1.4.1 *Ordinary Operation : (command and response)* 8
1.4.2 *Irregular operation : (command and response)* 8

2. COMMAND LIST:（INCLUDING STATUS AND ERROR CODE） 10

2.1.1 *Command list* 10
2.1.2 *Status code information format:* 13
2.2 E1, E0 ERROR CODE TABLE 13

3. COMMAND SPECIFICATION: 16

3.1 CARD DISPENSER OPERATION 16
3.1.1 *Reset (Initialization)* 16
3.1.2 *Status Request Command* 16
3.1.3 *Move card:* 17
3.1.4 *The card dispensing motor rotates backward* 18
3.1.5 *Front card entry command:* 18
3.2 AUTO-TEST TYPE OF IC CARD/RF CARD: 19
3.2.1 *Auto-test type of IC card:* 19
3.2.2 *Auto-test the type of RF card:* 19
3.3 CPU CARD OPERATION: 20
3.3.1 *CPU Card Reset (Activate):* 20
3.3.2 *CPU Card Power off:* 21
3.3.3 *Inquire CPU Card Status* 21
3.3.4 *T=0 protocol CPU Card APDU Operation* 21
3.3.5 *T=1 Protocol CPU Card APDU Operation* 22
3.3.6 *CPU Card Warm Reset* 23
3.3.7 *T=1, T=0 CPU Card Protocol Automatic Communication* 24
SAM(SECURE APPLICATION MODULE) CONTROL COMMAND 25
3.3.8 *Active SAM Command:* 25
3.3.9 *Deactivate SAM Command* 25
3.3.10 *Inquire SAM Status Command* 26
3.3.11 *SAM Communication T=0* 26
3.3.12 *SAM Communication T=1* 27
3.3.13 *SAM Warm Reset* 27
3.3.14 *Auto-Check SAM Card T=0/T=1 Protocol* 28

3.3.15 *Select SAM* *28*
3.4 SLE4442/4428 CARD OPERATION 29
3.4.1 *SLE4442/4428 card reset (activation):* *29*
3.4.2 *Deactivate SLE4442/4428 :* *29*
3.4.3 *Inquire status of SLE4442/4428:* *29*
3.4.4 *SLE4442 Control:* *30*
3.4.5 *SLE4428 control：* *34*
3.5 I2C MEMORY CARD OPERATION： 38
3.5.1 *Activate I2C memory card：* *38*
3.5.2 *Deactivate I2C memory card：* *39*
3.5.3 *Inquire Status of I2C memory card* *39*
3.5.4 *I2C operation：* *40*
3.6 CONTACTLESS IC CARD OPERATION： 42
3.6.1 *Activated contactless IC card：* *42*
3.6.2 *Deactivate RFID card：* *44*
3.6.3 *Inquire status of RFID card：* *44*
3.6.4 MIFARE 1 CARD CONTROL 45
3.6.5 *Type A RF card communication:* *57*
3.6.6 *Type B RFcard communication:* *57*
3.6.7 *Operations on cards using the 15693 protocol* *57*
3.6.8 *Read Serial number:* *61*
3.6.9 *Read information of machine configuration* *61*
3.6.10 *Error-card Bin Counter Control:* *61*
3.6.11 *Reset PC/SC reader (only available for CRT-591-MTX1)* *62*
3.7 GET FIRMWARE VERSION 62

Revisions record:

version	Date	content
1.0	02/07/2013	Initial release
1.1	2016.02.01	Add the card moving commands C2a,C2b,C2c,and C2d to the reader.
1.2	2021.01.13	Modify the ATR return for the activation of SLE4442/SLE4428 cards,changing SLE4442 to SLE4428.
1.4	2025.0528	1. Add support for cards such as UL C/AES. 2. The document is revised to the C591_MR series universal version communication protocol

1. Serial port control specification：

1.1.1 Communication format

Baud rate(BPS)： 9600/192000/38400/57600BPS（Automatically scan）

Communication method: Asynchronous

Transmission method： Half duplex, support multi-unit communication (16 units max)

Data frame structure：

Start bit	D0	D1	D2	D3	D4	D5	D6	D7	Stop sbit

Start bit：1 bit

Data length：8 bit

Check bit：none

Stop bit：1 bit

Character code： ASCII 8 bit code

1.1.2 Communication control method

Dispenser is a slave component, executes operation according to text(command) received from HOST.

Character reference

STX (F2H) Start of text

ETX (03H) End of text

ACK (06H) Acknowledge

NAK (15H) Negative acknowledge

EOT (04H) Cancel Clear Character

ADDR Address character

1.2 Communication format and control character

1.2.1 Send command package format

STX (0xF2)	ADDR (1 byte)	LENH (1 byte)	LENL (1 byte)	CMT (1 byte)	CM (1 byte)	PM (1 byte)	DATA (N bytes)	ETX (1 byte)	BCC (1 byte)

Range of TXET

Range of BCC calculation

MAX 1024 byte

STX (F2H) Start of text

LENH (1 byte) Length of high byte of text

LENL (1 byte) Length of low byte of text

CMT Command head (‘C’, 43H)

CM Specify as command

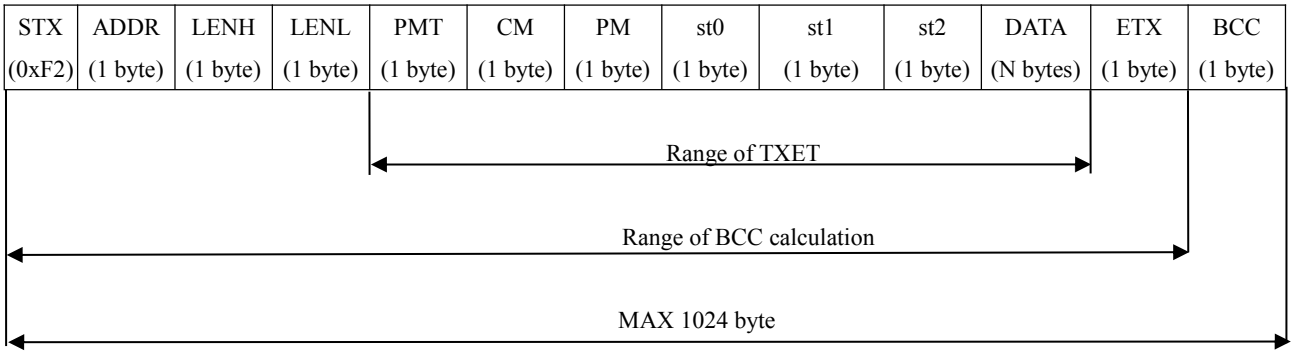
PM Command parameter

DATA Transmission data (N byte, N=0 ~ 512)

ETX (03H) End of text

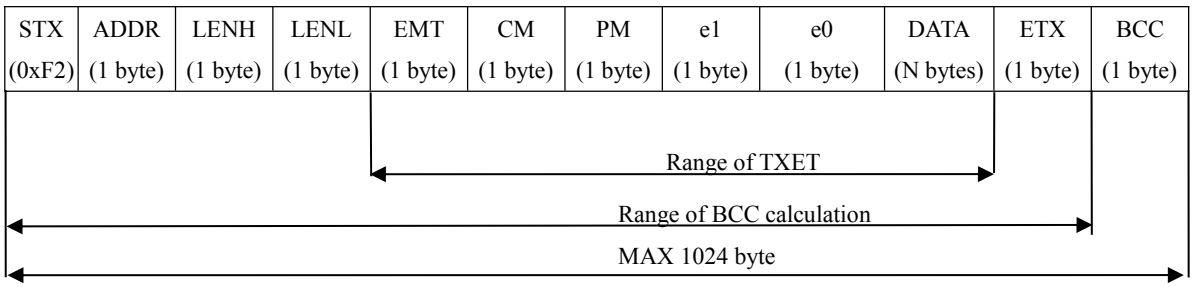
BCC (1 bytes) XOR

1.2.2 Successful responsive package format and character



STX (F2H)	Start of text
LENH (1 byte)	Length of high byte of text
LENL (1 byte)	Length of low byte of text
PMT	Return command head (‘P’, 50H)
CM	Specify as command
PM	Return command parameter
St0, st1, st2	Return dispenser status code
DATA	Return command data (N byte, N=0~512)
ETX (03H)	End of text
BCC (1 bytes)	XOR

1.2.3 Failed responsive package format and character



STX (F2H)	Start of text
LENH (1 byte)	Length of high byte of text
LENL (1 byte)	Length of low byte of text
EMT	Return command head (N’, 45H)
CM	Specify as command
PM	Return command parameter
e1, e0	Return dispenser error code
DATA	Return command data (N byte, N=0~512)
ETX (03H)	End of text
BCC (1 bytes)	XOR

1.3 Address set for multi-units communication：

ADDR(multi-units communication)：Address every unit by 4 dip switch on the main board：

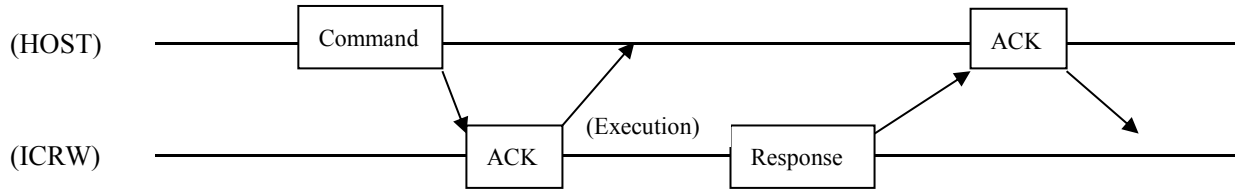
Address	DIP switch				ADDR
	4	3	2	1	
0#	ON	ON	ON	ON	00H
1#	ON	ON	ON	OFF	01H
2#	ON	ON	OFF	ON	02H
3#	ON	ON	OFF	OFF	03H
4#	ON	OFF	ON	ON	04H
5#	ON	OFF	ON	OFF	05H
6#	ON	OFF	OFF	ON	06H
7#	ON	OFF	OFF	OFF	07H
8#	OFF	ON	ON	ON	08H
9#	OFF	ON	ON	OFF	09H
10#	OFF	ON	OFF	ON	0AH
11#	OFF	ON	OFF	OFF	0BH
12#	OFF	OFF	ON	ON	0CH
13#	OFF	OFF	ON	OFF	0DH
14#	OFF	OFF	OFF	ON	0EH
15#	OFF	OFF	OFF	OFF	0FH

The default address for single device in ex-work is set as 00H, each device has unique address.

When conducting multi machine communication, each extension should be set as a unique address, and each extension should be selected for control through the corresponding address word in the communication package.

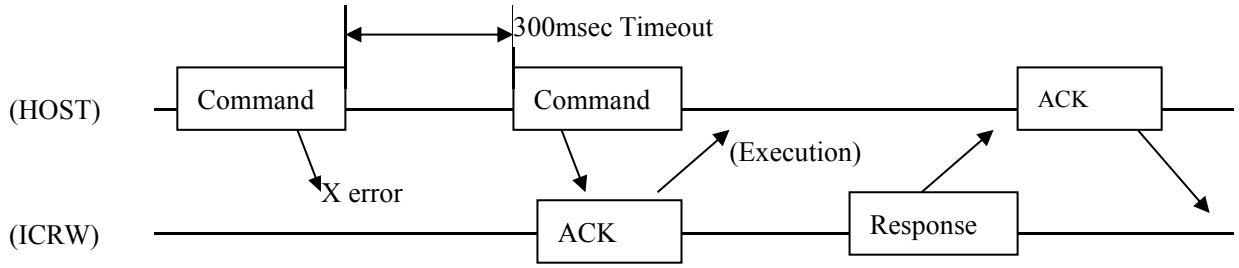
1.4 Communication method:

1.4.1 Ordinary Operation : (command and response)

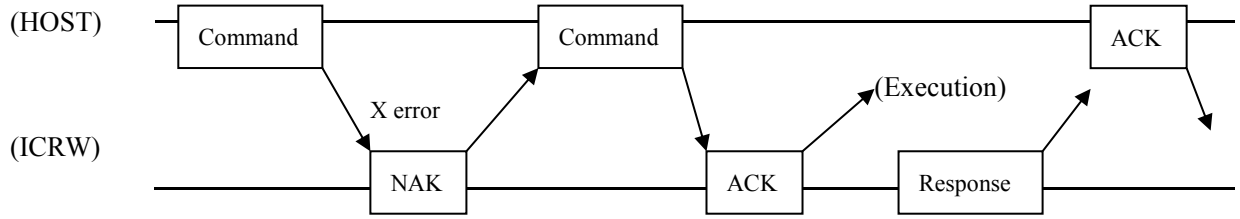


1.4.2 Irregular operation : (command and response)

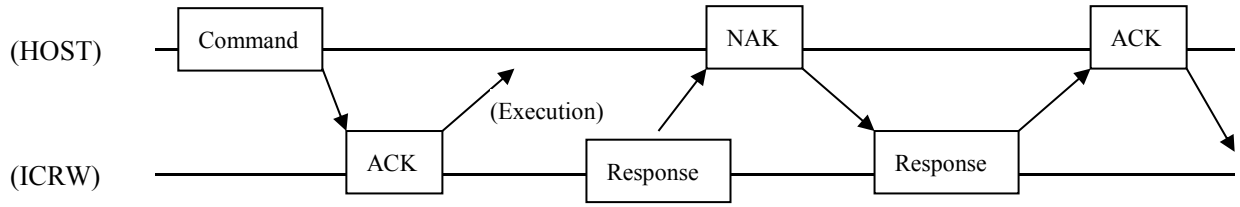
Case 1



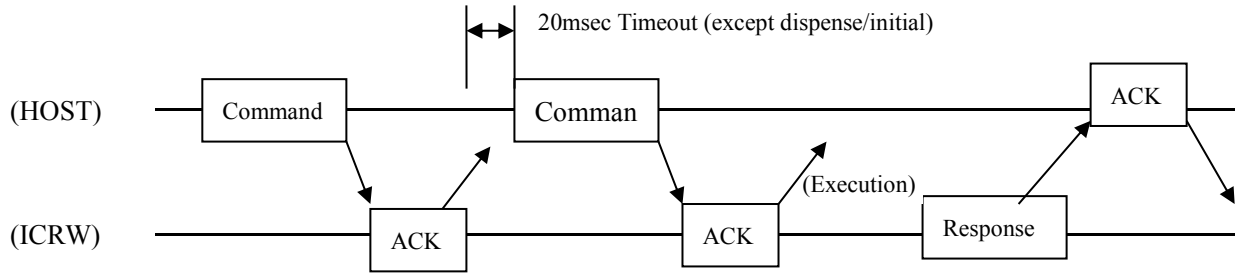
Case 2



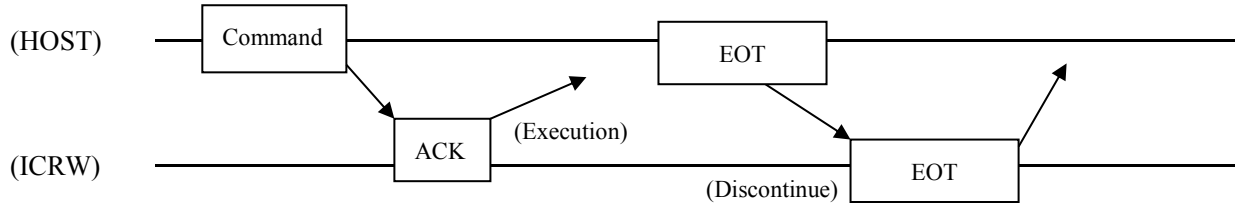
Case 3



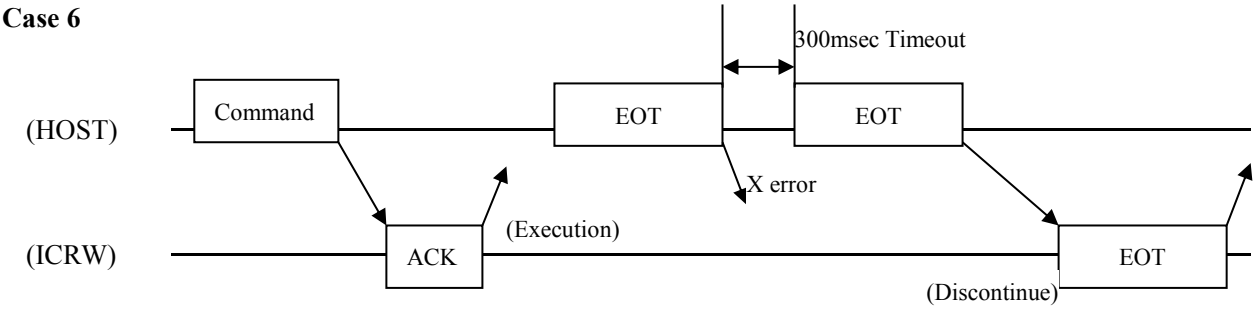
Case 4



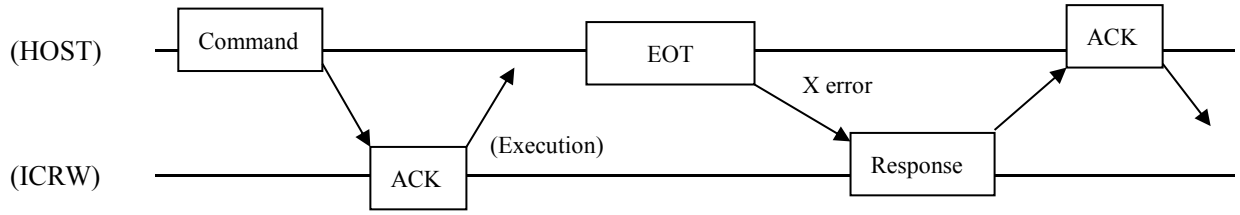
Case 5



Case 6



Case 7



2. Command list:（including status and error code）

2.1.1 Command list

Command	Function	CM	PM	description
INITIALIZE	Initialize CRT-591-M	30H	30H	If card is inside, move card to card holding position
			31H	If card is inside, capture card to error card bin
			33H	If card is inside, does not move the card.
			34H	Same as 30H and retract counter will work.
			35H	Same as 31H and retract counter will work.
			37H	Same as 33H and retract counter will work.
STATUS REQUEST	Inquire status	31H	30H	Report CRT-591M status
			31H	Report sensor status
CARD MOVE	Card movement	32H	30H	Move card to card holding positon
			31H	Move card to IC card position
			32H	Move card to RF card position
			33H	Move card to error card bin
			39H	Move card to gate
			61H	Move the card from the read-write area to the outlet (card holding position)
			62H	Move the card from the read-write area to the IC card position (only for card dispensing from card stack)
			63H	Move the card from the read-write area to the RF card position
			64H	Move the card from the read-write area to the outlet without card holding position
			72H	The card dispensing motor rotates backward
CARD ENTRY	From output gate	33H	30H	Enable card entry from output gate
			31H	Disable card entry from ouput gate
CARD TYPE	ICCard/RFCard TypeCheck	50H	30H	Autocheck ICCardType
			31H	Autocheck RFCardType
CPUCARD CONTROL	CPU Card Applicatio Opertion	51H	30H	CPUCard cold reset
			31H	CPUCard power down
			32H	CPUCard status check
			33H	T=0 CPUCard APDU data exchange

			34H	T=1 CPUCard APDU data exchange
			38H	CPUCard hot reset
			39H	Auto distinguish T=0/T=1 CPUCard APDU data exchange
SAM CARD CONTROL	SAMCard Application Operation	52H	30H	SAMCard cold reset
			31H	SAMCard down power
			32H	SAMCard status check
			33H	T=0 SAMCard APDU data exchange
			34H	T=1 SAMCardAPDU data exchange
			38H	SAMCard hot reset
			39H	Auto distinguish T=0/T=1 SAMCardAPDU data exchange
			40H	Choose SAMCard stand
SLE4442/4428CARD CONTROL	SLE4442/4428CARD CONTROL	53H	30H	SLE4442/4428Card reset
			31H	SLE4442/4428Card power down
			32H	Browse SLE4442/4428Card status
			33H	Operate SLE4442Card
			34H	Operate SLE4428Card
IC MEMORY CARD	24C01—24C256Card Operation	54H	30H	ICCard reset
			31H	ICCard down power
			32H	Check ICCard status
			33H	Read ICCard
			34H	Write ICCard
RFCARD CONTROL (13.56 MHZ)	Mifare standard card Type A & B T=CL protocol operation	60H	30H	RF Card startup
			31H	RF Card down power
			32H	RF Card operation status check
			33H	Mifare standard Card read/write
			34H	Type A standard T=CLCard APDU data exchange
			35H	Type B standard T=CLCard APDU data exchange
			39H	RF card wake-up/sleep
Read machine serial number	Read machine serial number	A2H	30H	Read machine Serial number
Read machine CONFIG	Read machine CONFIG	A3H	30H	Read machine configuration information
RECYCLE BIN COUNTER	RECYCLEBIN COUNTER	A5H	30H	Read number of counter of Card error card bin
			31H	Set Initiate value of error card bin

				counter
Perform a reset on the PC/SC card reader.	Perform a reset on the PC/SC card reader.	A6H	30H	Perform a reset on the PC/SC card reader.
READ FIRMWARE		DCH	31H	Read firmware version information

2.1.2 Status code information format：

Card Status Code (st0,st1,st2）

st0	**Content**
“0”	No Card in CRT-591M
“1”	One Card in gate
“2”	One Card on RF/IC Card Position

st1	**Content**
“0”	No Card in stacker
“1”	Few Card in stacker
“2”	Enough Cards in card box

st2	**Content**
“0”	Error card bin not full
“1”	Error card bin full

2.2 e1, e0 Error Code Table

e1,e0 code	**Content**
“00”	Reception of Undefined Command
“01”	Command Parameter Error
“02”	Command Sequence Error
“03”	Out of Hardware Support Command
“04”	Command Data Error (error in communication package data)
“05”	IC Card Contact Not Release
“06”	
“07”	
“08”	
“09”	
“10”	Card Jam
“11”	
“12”	sensor error
“13”	Too Long-Card
“14”	Too Short-Card
“15”	
“16”	Card position movement
“17”	
“18”	
“19”	
“20”	
“21”	

“22”	
“23”	
“24”	
“25”	
“26”	
“27”	
“28”	
“29”	
“30”	
“31”	
“32”	
“33”	
“34”	
“35”	
“36”	
“37”	
“38”	
“39”	
“40”	Disability of Recycling card (When the card is recycled, it gets pulled away)
“41”	Soleniod for IC Card Error
“42”	
“43”	Disable To Move Card To IC Card Position
“44”	
“45”	Manually Move Card
“46”	
“47”	
“48”	
“49”	
“50”	Received Card Counter Overflow
“51”	Motor error
“52”	
“53”	
“54”	
“55”	
“56”	
“57”	
“58”	
“59”	
“60”	Short Circuit of IC Card Supply Power
“61”	Activiation of IC Card failure
“62”	Command Out Of IC Card Support
“63”	Transmission of IC card data error
“64”	
“65”	Un-activate of IC Card

“66”	Command Out Of IC Current Card Support
“67”	IC Card Transmittion Error
“68”	IC Card transmittion Overtime
“69”	CPU/SAM Non-Compliance To EMV Standard
“A0”	Empty-Stack, no card in the card stack
“A1”	Error card bin full
“A2”	
“A3”	
“A4”	
“A5”	
“A6”	
“A7”	
“A8”	
“A9”	
“B0”	Not Reset

3. Command Specification:

3.1 Card dispenser operation

3.1.1 Reset (Initialization)

HOST Command (TXET):

“C”	30H	Pm

Positive response (TXET):

“P”	30H	Pm	st0	st1	st2	Rev_type

Negative response (TEXT):

“N”	30H	Pm	e1	e0

This command should be received before any other command after card dispenser is power on, otherwise card dispenser can not execute any other commands; Card dispenser will auto detect ICRW and Host’s BAUD and adapt corresponding BAUD when it receive this command for the first time. On receiving this command, card dispenser will clear all error code, Disable card entry from ouput gate and return FW version.

Pm: Command parameter

If there is no card in card dispenser, engine will rotate slightly to clear up card in stacker

If there are cards in card dispenser, the disposal is show as below:

=30HMove the card to Gate portion

=31HCapture card to reject-stacker

=33HIf card is inside card dispenser, does not move the card

=34HSame as pm=30H, and Retract counter will work

=35HSame as pm=31H, and Retract counter will work

=37HSame as pm=33H, and Retract counter will work

Rev_type: FW version, “CRT-591-M001”

3.1.2 Status Request Command

HOST Command

“C”	31H	Pm

Positive response

“P”	31H	Pm	st0	st1	st2	Sensor(10 byte)

Negative response

“N”	31H	Pm	e1	e0

Pm=30H Report current status (see 2.2 for more detail)

Pm=31H Report current status and sensor status (8 Bytes) as following table:

Sensor	status	status
S1	30H No card	31H With card
S2	30H No card	31H With card
S3	30H No card	31H With card
S4	30H No card	31H With card
S5	30H No card	31H With card
S6	30H No card	31H With card
S7	30H No card	31H With card
S8	Reserve	

3.1.3 Move card:

HOST command:

"C"	32H	Pm

Positive response:

"P"	32H	Pm	st0	st1	st2

Negative response:

"N"	32H	Pm	e1	e0

Pm=30H move card to card holding position

Pm=31H move card to IC card read/write position (dispense from stacker)

Pm=32H move card to RF read/write position

Pm=33H retract card to error card bin

Pm=39H move card out of gate

Pm=61H (a) Move the card from the read-write area to the outlet (card holding position).

Pm=62H (b) Move the card from the read-write area to the IC card position (only for card dispensing from card stack)

Pm=63H (c) Move the card from the read-write area to the RF card position

Pm=64H (d) Move the card from the read-write area to the outlet without card holding position

If card can not move to specified position, CRT-591-M will return Card jam error

Note: When execute Capture card command, if error card bin is full, CRT-591M will return error card bin error

3.1.4 The card dispensing motor rotates backward

HOST command:

"C"	32H	Pm	Time(1 byte)

Card feeding successful return:

"P"	32H	Pm	st0	st1	st2

Failed to feed card:

"N"	32H	Pm	e1	e0

Pm = 72H (r) The card dispensing motor rotates backward.(By default,it rotates backward for 200ms.This can be parameterized with the parameter Time,which takes values from'1'to'9',representing 100ms to 900ms.)

This action pulls the card that is stopped at the card box exit back into the card box to prevent the card from obstructing the removal of the card box when it is extracted(ignore if there is no card box).

3.1.5 Front card entry command:

HOST command:

"C"	33H	Pm

Positive response:

"P"	33H	Pm	st0	st1	st2

Negative response:

"N"	33H	Pm	e1	e0

After set card input from gate available, if insert card from gate, CRT-591-M will carry the card to RF Card operation position.

Pm=30H Enable card input from gate

Pm=31H Disable card input from gate

Note: Execute reset command, CRT-591-M will disable card input from gate.

3.2 Auto-Test type of IC card/RF card:

3.2.1 Auto-test type of IC card:

HOST command:

“C”	50H	30H

Positive response:

“P”	50H	30H	st0	st1	st2	Card_type

Negative response:

“N”	50H	30H	e1	e0

Auto-test current type of IC card, move the card of inner machine to IC card position, auto-test the current type of IC card, Return the information of Card_type after finishing testing.

Cart_type(2 byte)		Instruction
‘0’	‘0’	Unknown IC card_type
‘1’	‘0’	T=0 CPU
	‘1’	T=1 CPU
‘2’	‘0’	SLE4442
	‘1’	SLE4428
‘3’	‘0’	AT24C01
	‘1’	AT24C02
	‘2’	AT24C04
	‘3’	AT24C08
	‘4’	AT24C16
	‘5’	AT24C32
	‘6’	AT24C64
	‘7’	AT24C128
	‘8’	AT24C256

3.2.2 Auto-test the type of RF card:

HOSTcommand:

“C”	50H	31H

Positive response:

“P”	50H	31H	st0	st1	st2	Card_type

Negative response:

“N”	50H	31H	e1	e0

Auto-test current type of RF card, move the card of inner machine to RF card position, auto-test the current type of RF card, Return the information of Card_type after finishing testing.

Cart_type(2 byte)		Instruction
‘0’	‘0’	Unknown RF card_type
‘1’	‘0’	Mifare one S50
	‘1’	Mifare one S70
	‘2’	Mifare one UL
‘2’	‘0’	Type A CPU
‘3’	‘0’	Type B CPU

3.3 CPU card operation:

3.3.1 CPU Card Reset (Activate):

HOST Command:

“C”	51H	30H	Vcc

Positive response:

“P”	51H	30H	st0	st1	st2	Type	ATR

Negative response:

“N”	51H	30H	e1	e0	Type	ATR

To cold reset IC card. The ICRW supplies power (VCC) and clock (CLK) , Card activated, return ATR.

Vcc=30H: CRT-591M supplies with +5V to VCC and activates in line with the EMV2000 ver4.0.

Vcc=33H: CRT-591M supplies with +5V to VCC and activates in line with the ISO/IEC7816-3.

Vcc=35H: CRT-591M supplies with +3V to VCC and activates in line with the ISO/IEC7816-3.

In case there is no Vcc word, it will have 30H as default value.

During the reset process, If ATR is not compliance to EMV, return e1,e0=“69”

During the reset process, the detection of the IC card power supply failed with an error of e1, e0="60"

Type: CPU Card protocol Type

=30H T=0 protocol CPU Card

=31H T=1 protocol CPU Card

Format of ATR: the format of the reset response information is as follows:

TS	TO	TA1	TB1	…	TCK

3.3.2 CPU Card Power off:

HOST Command:

“C”	51H	31H

Positive response:

“P”	51H	31H	st0	st1	st2

Negative response:

“N”	51H	31H	e1	e0

Power off operation to CPU Card.

Power off operation to power on and activated CPU card.

3.3.3 Inquire CPU Card Status

HOST Command:

“C”	51H	32H

Positive response:

“P”	51H	32H	st0	st1	st2	Sti

Negative response:

“N”	51H	32H	e1	e0

ICRW tells the status of IC card with sti:

St i=30H Card not activated

=31H Card have activated, current CPU Card working frequency is 3.57 MHZ

=32H Card have activated, current CPU Card working frequency is 7.16 MHZ

If IC Card power error, return e1,e0=“60”

3.3.4 T=0 protocol CPU Card APDU Operation

HOST Command:

“C”	51H	33H	C-APDU

Positive response:

“P”	51H	33H	st0	st1	st2	R-APDU

Negative response:

“N”	51H	33H	e1	e0

Perform data exchange operation on the T=0 protocol CPU card that has been successfully reset (activated).

C-APDU HOST sends data packets to the T=0 CPU card, with a minimum of 4 bytes and a maximum of 261 bytes. The format is as follows:

CLA	INS	P1	P2	LC	Data1	∘∘∘∘	Le

R-APDU CPU card returns a HOST data packet, with a minimum of 2 bytes and a maximum of 258 bytes. The format is as follows:

Data1	…..	Data(n)	Sw1	Sw0

If the IC card power supply fails to be detected during the operation, the card machine returns an e1, e0-="60" error;

If the current CPU card of the protocol is not a T=0 protocol, the card machine returns error code e1, e0="62" error;

If the CPU card times out, the card machine will first power down the CPU card and then return e1, e0="63" error;

If other protocol errors occur, the card will first power down the CPU card and then return error code e1, e0="64" error;

If HOST does not perform reset activation operation in SAM, returns e1, e0="65" error

Note: Please refer to ISO/IEC7816-3 for the format of T=0 APDU. The specific C-APDU command operation is based on the card's COS instruction.

3.3.5 T=1 Protocol CPU Card APDU Operation

HOST Command:

“C”	51H	34H	C-APDU

Positive response:

“P”	51H	34H	st0	st1	st2	R-APDU

Negative response:

“N”	51H	34H	e1	e0

Perform data exchange operations on the T=1 protocol CPU card that has been successfully reset (activated).

The card reader performs the T=1 protocol C-APDU operation, combining the C-APDU sent by HOST into an I-bolck according to the T=1 protocol specification, and sending it to the CPU card. Then, the I-block returned by the CPU card extracts the R-ADPU packet and returns it to HOST. The process is as follows:

A. Send C-APDU (add block header and block tail, combine the sent C-APDU as block information, and send it to the CPU card as an I-block).

C-APDU

CLA	INS	P1	P2	Lc	Data1	…	Data(Lc)	Le

↓

NAD	PCB	LEN	CLA	INS	P1	P2	Lc	Data1	…	Data(Lc)	Le	EDC

	Head block	Information block	End block
I-block			

B. Receive R-APDU (return I-block from CPU card, return block information to R-ADPU)

	Head block			Information block									End block
I-block	NAD	PCB	LEN	CLA	INS	P1	P2	Lc	Data1	…	Data(Lc)	Le	EDC

↓

R-APDU	CLA	INS	P1	P2	Lc	Data1	…	Data(Lc)	Le

If the IC card power supply fails to be detected during the operation, returns an e1, e0-="60" error;

If the current CPU card of the protocol is not the T=1 protocol, returns error code e1, e0="62" error;

If the CPU card times out, will first power down the CPU card and then return e1, e0="63" error;

If other protocol errors occur, will first power down the CPU card and then return error code e1, e0="64" error;

If HOST does not perform reset activation operation in SAM, returns e1, e0="65" error

Please refer to ISO/IEC7816-3 for the format of T=1 APDU. The specific C-APDU command operation is based on the card's COS instruction.

3.3.6 CPU Card Warm Reset

HOST Command:

“C”	51H	38H

Positive response:

“P”	51H	38H	st0	st1	st2	Type	ATR

Negaitive response:

“N”	51H	38H	e1	e0

Keeping the status of the IC contact activated, then returns response upon receiving “ATR” again.

Type: CPU Card communication protocol

=30H T=0 Protocol

=31H T=1 Protocol

3.3.7 T=1, T=0 CPU Card Protocol Automatic Communication

HOST Command:

“C”	51H	39H	C-APDU

Positive response:

“P”	51H	39H	st0	st1	st2	R-APDU

Negative response:

“N”	51H	39H	e1	e0

The machine automatically determines the current CPU card T=0/T=1 protocol, automatically selects the C-APDU operation for the corresponding protocol, and returns to R-APDU.

During the operation, the detection of the IC card power supply failed, and returned an e1, e0-="60" error;

If the current CPU card of the protocol is not T=0/T=1 protocol, returns error code e1, e0="62" error;

If the CPU card times out, will first power down the CPU card and then return e1, e0="63" error;

If other protocol errors occur, will first power down the CPU card and then return error code e1, e0="64" error;

If HOST does not perform reset activation operation in SAM, returns e1, e0="65" error.

SAM(Secure Application Module) Control Command

3.3.8 Active SAM Command:

HOSTCommand:

"C"	52H	30H	Vcc

Positive response:

"P"	52H	30H	st0	st1	st2	Type	ATR

Negative response:

"N"	52H	30H	e1	e0	Type	ATR

The CRT-591 supplies power (VCC) and clock (CLK), then reset (RST) release.

Type: SAM protocol type

=30H T=0 protocol

=31H T=1 protocol

ATR(Answer To Reset) format:

TS	TO	TA1	TB1	…	TCK

Vcc=30H: ICRW supplies with +5V to VCC and activates in line with the EMV2000 ver4.0.

Vcc=33H: ICRW supplies with +5V to VCC and activates in line with the ISO/IEC7816-3.

Vcc=35H: ICRW supplies with +3V to VCC and activates in line with the ISO/IEC7816-3.

Incase there is no Vcc, it will have 30H as default value

If ATR is not compliance to EMV, return e1,e0="69"

Notes : There will be error and return ATR & Type when reset in line with EMV return

When a power failure is recognized while a power supply is supplied to the card, error code e1,e0="60" is returned.

3.3.9 Deactivate SAM Command

HOST Command:

"C"	52H	31H

Positive response:

"P"	52H	31H	st0	st1	st2

Negative response:

"N"	52H	31H	e1	e0

This deactivates SAM

Power off operation to power on and activated SAM card.

3.3.10 Inquire SAM Status Command

HOST Command:

“C”	52H	32H

Positive response:

“P”	52H	32H	st0	st1	st2	Sti	Stj

Negatiive response:

“N”	52H	32H	e1	e0

CRT-591-M return the status of SAM with sti. stj

Sti =30H SAM is deactivated

Sti =31H SAM is activated, working frequency is 3.57 MHZ

Sti =32H SAM is activated, working frequency is 7.16 MHZ

Stj =30H First SAM card connector

Stj =31H Second SAM card connector (Optional)

Stj =32H Third SAM card connector (Optional)

Stj =33H Fourth SAM card connector(Optional)

Stj =34H Fifth SAM card connector(Optional)

An error e1,e0=“60”is returned when a power failure is detected.

3.3.11 SAM Communication T=0

Command

“C”	52H	33H	C-APDU

Positive response:

“P”	52H	33H	st0	st1	st2	R-APDU

Negative response:

“N”	52H	33H	e1	e0

Perform data exchange operation on the T=0 protocol SAM card that has been successfully reset (activated).

If protocol type of IC card is not T=0, error code “62”is sent.

If ICC responds time out, CRT-591-M deactivates an IC card and error code “63” is sent.

If any other protocol error occurs, CRT-591-M deactivates an IC card and error code “64” is sent.

If HOST tries to communicate before an IC card activation, error code “65” is sent.

Note: If you want to more about T=0 APDU format. Please refer to ISO/IEC7816-3 and COS command

3.3.12 SAM Communication T=1

Command

“C”	52H	34H	C-APDU

Positive response:

“P”	52H	34H	st0	st1	st2	R-APDU

Negative response:

“N”	52H	44H	e1	e0

Perform data exchange operations on the T=1 protocol SAM card that has been successfully reset (activated).

If the IC card power supply fails to be detected during the operation, the machine returns an e1, e0-="60" error;

If the current SAM card in the protocol is not a T=1 protocol, the machine returns error code e1, e0="62" error;

If the SAM card times out, the machine will first power down the SAM card and then return an e1, e0="63" error;

If other protocol errors occur, the SIM card will first power down the SAM card and then return error code e1, e0="64" error;

If HOST does not perform reset activation operation in SAM, returns e1, e0="65" error

Please refer to ISO/IEC7816-3 for the format of T=1 APDU. The specific C-APDU command operation is based on the COS instruction of the card

3.3.13 SAM Warm Reset

Command

“C”	52H	38H

Positive response:

“P”	52H	38H	st0	st1	st2	Type	ATR

Negative response:

“N”	52H	38H	e1	e0

Keeping the status of the SAM activated, then returns response upon receiving.

Type: SAM protocol type

=30H T=0 Protocol

=31H T=1 Protocol

3.3.14 Auto-Check SAM Card T=0/T=1 Protocol

Command

“C”	52H	39H	C-APDU

Positive response

“P”	52H	39H	st0	st1	st2	R-APDU

Negative response

“N”	52H	39H	e1	e0

The machine automatically determines the current SAM card T=0/T=1 protocol, automatically selects the C-APDU operation for the corresponding protocol, and returns to R-APDU.

During the operation, the detection of the IC card power supply failed, and returned an e1, e0-="60" error;

If the current SAM card in the protocol is not T=0/T=1 protocol, the machine returns error code e1, e0="62" error;

If the SAM card times out, the machine will first power down the SAM card and then return an e1, e0="63" error;

If other protocol errors occur, the SIM card will first power down the SAM card and then return error code e1, e0="64" error;

If HOST does not perform reset activation operation in SAM, returns e1, e0="65" error

3.3.15 Select SAM

Command

“C”	52H	40H	SAMn

Positive response

“P”	52H	40H	st0	st1	st2

Negative response

“N”	52H	40H	e1	e0

Select the SAM card on the SAM board

SAMn = 30H: SAM 1.

= 31H: SAM 2. (option)

= 32H: SAM 3. (option)

= 33H: SAM 4. (option)

= 34H: SAM 5. (option)

This command is only valid for models with PSAM cards and can only be operated on one SAM card at a time. SAM1 is automatically selected during card initialization.

3.4 SLE4442/4428 card operation

3.4.1 SLE4442/4428 card reset (activation):

HOST command:

“C”	53H	30H

Positive response:

“P”	53H	30H	st0	st1	st2	ATR(4 byte)

Negative response:

“N”	54H	30H	e1	e0

The CRT-591-M supplies power (VCC), clock(CLK)，then reset(RST),after reset，return ATR.

ATR： SLE4442 Card ATR=“A2H，13H，10H，91H”

SLE4442 Card ATR=“92H，23H，10H，91H”

3.4.2 Deactivate SLE4442/4428 :

HOST command :

“C”	53H	31H

Positive response :

“P”	53H	31H	st0	st1	st2

Negative response :

“N”	53H	31H	e1	e0

The CRT-591-M stop suppling power (VCC), clock(CLK)，and then reset(RST) release.

3.4.3 Inquire status of SLE4442/4428:

HOST command :

“C”	53H	32H

Positive response :

“P”	53H	32H	st0	st1	st2	Sti

Negative response :

“N”	54H	32H	e1	e0

This command is used to inquire the status of card ,will return Sti after the command successfully execute.

Sti= 30H SLE4442/4428 Deactivated

Sti= 31H SLE4442 Activated

Sti= 32H SLE4428 Activated

3.4.4 SLE4442 Control:

These functions are specified by a command data form like C-APDU which format is based ISO/IEC 7816 T=0 standard.

In this case, the CRT-591-M recognizes the meaning of the command data,and execute the treatment related to the card by controlling hardware.

After the command was executed properly,CRT-591-M returns a positive response with response data 9000H; When an error occurs during the communication ,the CRT-591-M returns a positive response with status information in response data“sw1+sw2”which is base on ISO/IEC 7816-3 T=0 standard.

Sw1	Sw2	Specification
90H	00H	success
6FH	00H	fail
6FH	01H	Key validation error
6FH	02H	Key validation error and lock
67H	00H	Address overflow
6BH	00H	Operation length overflow

3.4.4.1 Data read from main memory on SLE4442：

HOST command:

“C”	53H	33H	00H	B0H	00H	abH	cdH

Positive response:

“P”	53H	33H	st0	st1	st2	data

Negative response:

“N”	53H	33H	e1	e0

Notes： ab: the start address to read data in the main memory

cd: the length of bytes of data to read

CRT-591-M read data from the main memory of SLE4442,and transmits data on abH and cdH . The capacity of the main memory is 256 byte. All the contents of the main memory can be with the following commandEx). “CR3”+00B0000000

3.4.4.2 Data read from protection memory on SLE4442：

HOST command:

“C”	53H	33H	00H	B0H	01H	abH	cdH

Positive response:

“P”	53H	33H	st0	st1	st2	data

egative response:

“N”	53H	33H	e1	e0

Notes： ab: the start address to read data in the main memory

cd: the length of bytes of data to read

SLE4442 card of all 32 bits data in the protection memory as the data on 4 bytes. Corresponding protection address is 00H—1FH。All the SLE442 of protection memory can be read with the following command Ex). “CR3”+00B0010004

3.4.4.3 Data read from security memory on SLE4442

HOST Command:

“C”	53H	33H	00H	B0H	02H	abH	cdH	efH…

Positive response:

“P”	53H	33H	st0	st1	st2	data

Negative response:

“N”	53H	33H	e1	e0

Note： ab: the start address to read data in the main memory

cd: the length of bytes of data to read

To read the security data of SLE4442 card.

SLE4442 card with 4bytes security area， 1 byte code error data and 3 byte password values (the password data can be read after correct check password). All the SLE442 card security area data can be read with the following command

Ex). “CR3”+00B0020004

3.4.4.4 Data write main memory on SLE4442:

HOST command:

“C”	53H	33H	00H	D0H	00H	abH	cdH	efH…

Positive response:

“P”	53H	33H	st0	st1	st2	data

Negative response:

“N”	53H	33H	e1	e0

Notes： ab: The start address to write data in the main memory

cd: the length of bytes of data to write

ef: the data to write first (cdH byte)

To read the data on SLE4442 main memory，write the appointed data in main memory，will return the operating results after the CRT-591-M has been wrote the date with validation. Before write the main memory,must validate the SLE442 password.

The capacity of the main memory is 256 byte, the byte number cd=00H of data to write means 256bytes.

The example that data are written in the whole area of the main memory is shown in the following.

Ex). “CR3”+00D000000+Write data(256 byte)

After command execution, CRT-591-M returns response with 9000H or sw1, sw2 as the result.

If the addressed data on main memory is protected by the protect status, data is not allow.

Data write to protection memory on SLE4442

HOST command:

“C”	53H	33H	00H	D0H	01H	abH	cdH	efH…

Positive response :

“P”	53H	33H	st0	st1	st2	data

Negative response:

“N”	53H	33H	e1	e0

notes： ab: The start address to write data in the main memory

cd: the length of bytes of data to write

ef: the data to write first (cdH byte)

TO write protect unit of memory for main memory of protection .Before execute the command， much to validate the SLE442 card password. The address of the main memory that the protection is possible is 00H—1FH。Each protection condition of the 00H—1FH main memory can be controlled with 32 bit in the protection memory. if byte0 of the protection memory byte0 is “1” ,data on the address 00H of the main memory are protected.

The content of protect status can not be change once setting protection.

For example:write 10H data to 20H address and set up protection

Ex). “CR3”+00D001100120

After command execution ,CRT-591-M return with 9000H (successful) or sw1,sw2(fail) as the result.

CRT-591-M readd data first from the main memory ,and it is compared with the value that it was received.When this is wrong ,writing is not begun.

Protection condition can be set up at one time in the data which continued in the main memory.

3.4.4.5 Data write to security memory on SLE4442:

Host Command

“C”	53H	33H	00H	D0H	02H	abH	cdH	efH…

Positive response

“P”	53H	33H	st0	st1	st2	data

Negative response

“N”	53H	33H	e1	e0

Notes: ab H : the start address to write data in the main memory

cd H : the length of bytes of data to write

ef H : the data to write first (cd H bytes)

After a password check is finished normally, the Reference-Data area of 3byte can be changed.

All 32bits are handled as 4bytes. How to change the Reference-Data is as the following.

ex). "CR3"+ 00D0020103123456

After command execution, ICRW returns response with 9000H or sw1+sw2 as the result.

Notes: Better not ot writ, because the Error-counter is always allowed to write and easily make a failure. Error-Counter is controlled when password is checked。

3.4.4.6 Verification data present to SLE4442:

Host Command

“C”	53H	33H	00H	20H	03H	01H	03H	efH…

Positive response

“P”	53H	33H	st0	st1	st2	data

Negative response

“N”	53H	33H	e1	e0

Notes : ef H : the data to compare (3bytes)

Before changing data, password must be check

Because this function should be made effective, the issue of the next command is necessary.

Ex). “CR3”+0020030103xxxxxx (xxxxxx : security code 3bytes)

Card will verify password between card and command.

A user must know password at least when a user wants to rewrite the data on SLE4442 card. Error-Counter can be reset in the zero if password is given to SLE4442 card properly if the value of Error-Counter is 2 or less

3.4.5 SLE4428 control：

These functions are specified by a command data form like C-APDU which format is based on T=0 standard.

In this case, CRT-591M recognizes the meaning of the command data, and execute the treatment related to the card by controlling hardware.

After the command was executed properly, CRT-591M returns a positive response with response data 9000H like from the IC card. When an error occurs during the communication with SLE4442, CRT-591M returns a positive response with status information in response data "sw1+sw2” which is base on ISO/IEC 7816-3

Sw1	Sw2	Specification
90H	00H	Success
6FH	00H	Fail
6FH	01H	Key Validation error
6FH	02H	Key Validation error and Lock
6BH	00H	Address overflow
67H	00H	Operation length overflow

3.4.5.1 Data Reading of main-memory of SLE4428：

Host Command

“C”	53H	34H	00H	B0H	0aH	bcH	deH

Positive response

“P”	53H	34H	st0	st1	st2	data

Negative response

“N”	53H	34H	e1	e0

Notes: abc H : the start address to read data in the main memory

de H : the number of bytes of data to read

CRT-591-M read data from main memory of SLE4428 through abcH and deH

The capacity of the main memory is 1024bytes.

De="00"

Data to read means 256bytes.

The head part of the main memory can be read with the following command.

ex). "CR4"+00B0000000

3.4.5.2 Reading of protection-bit of SLE4428：

Host Command

“C”	53H	34H	00H	B0H	10H	abH	cdH

Positive response

“P”	53H	34H	st0	st1	st2	data

Negative response

“N”	53H	34H	e1	e0

Notes: ab H : read the start address to read the image of protection data of the main memory

cd H : read data operation length

The protection conditions of 1024bytes of main-memory are changed into the data on 1024bits, and it is read.

1024bits is equivalent to 128bytes. (1024 = 128 x 8)

Data to read first become protection information to address (000H-007H) of main-memory in the case of abH=00H.

The contents of the whole protection image can be read with the following command.

ex). "CR4"+00B0100080

The device read protection-bit of SLE4428 according to abH.

3.4.5.3 Data writing to main-memory of SLE4428：

Host Command

“C”	53H	34H	00H	D0H	0aH	bcH	deH	fgH…

Positive response

“P”	53H	34H	st0	st1	st2	data

Negative response

“N”	53H	34H	e1	e0

Notes: abc H : the start address to write data in the main memory

de H : the number of bytes of data to write

fg H : the data to write first (de H bytes)

The device writes data in the main memory. it returns a result after written data are checked.

Before doing this operation, password check must be done

The capacity of the main memory is 1024 bytes.

The example that data are written in from the address 100H is shown in the following.

ex). "CR4"+ 00D0010000 + Write Data (256byte)

After command execution, ICRW returns response with 9000H or sw1+sw2 as the result.
If the addressed data on main memory is protected, the write operation is not available。

3.4.5.4 Verification of SLE4428 with protecting：

Host Command

“C”	53H	34H	00H	D0H	1aH	bcH	deH	fgH…

Positive response

“P”	53H	34H	st0	st1	st2	data

Negative response

“N”	53H	34H	e1	e0

Notes: abc H : the start address to write data in the main memory

de H : the number of bytes of data to write

fg H : the data to write first (de H bytes)

The device writes data in the main memory. It returns a result after written data are checked
Before doing this operation, password check must be done

3.4.5.5 Written with protection-bit：

Host Command

“C”	53H	34H	00H	D0H	2aH	bcH	deH	fgH…

Positive response

“P”	53H	34H	st0	st1	st2	data

Negative response

“N”	53H	34H	e1	e0

Notes: abc H : the start address to write data in the main memory

de H : the number of bytes of data to write

fg H : the data to write first (de H bytes)

Before doing this operation that writing data with protection-bit, password check must be done

After command execution, ICRW returns response with 9000H or sw1+sw2 as the result.

The device reads data first from the main memory, and it is compared with the value that it was received. When this is wrong, writing isn't begun. Protection condition can be set up at a time in the data which continued in the main memory。

3.4.5.6 Verification of password present to SLE4428:

Host Command

“C”	53H	34H	00H	20H	00H	00H	02H	efH…

Positive response

“P”	53H	34H	st0	st1	st2	data

Negative response

“N”	53H	34H	e1	e0

Notes: ef H : the data to compare (2bytes)

Before changing data, Password must be checked properly with SLE4428.

Because this function should be made effective, the issue of the next command is necessary.

ex). "CR4"+ 0020000002xxxx (xxxx : security code 2bytes)

The presented data are compared with internal data in SLE4428 card itself.

User should know the password of cad if they want to change the data in SLE4442, Error-Counter can be reset in the zero from 7 or less than 7. when error-counter is reset as zero, lock the card

3.5 I2C Memory Card operation：

3.5.1 Activate I2C memory card：

Host Command

“C”	54H	30H	Wrd	Vcc

Positive response

“P”	54H	30H	st0	st1	st2

Negative response

“N”	54H	30H	e1	e0

To activate (24C01,24C02,24C04,24C08,24C16,24C32,24C64,24C128,24C256) card

Device supplies a power supply (Vcc), Clock(CLK), Reset(RST).

Including:

Wrd set I2C type

Wrd =30 H To activate(24C01,24C02,24C04,24C08,24C16,24C32,24C64,24C128,24C256) card

Wrd =31 H To activate 24C01card

Wrd =32 H To activate 24C02 card

Wrd =33 H To activate 24C04 card

Wrd =34 H To activate 24C08 card

Wrd =35 H To activate 24C16 card

Wrd =36 H To activate 24C32 card

Wrd =37 H To activate 24C64 card

Wrd =38 H To activate 24C128 card

Wrd =39 H To activate 24C256 card

Vcc choose voltage to card

Vcc=30H 5V

Vcc=31H 3V

Vcc is optional parameter, no Set parameter in command is equal to Set=30H

3.5.2 Deactivate I2C memory card：

Host Command

“C”	54H	31H

Positive response

“P”	54H	31H	st0	st1	st2

Negative response

“N”	54H	31H	e1	e0

The device stops supplying a power supply (Vcc), Clock(CLK), Reset(RST)

3.5.3 Inquire Status of I2C memory card

HOST Command

“C”	54H	32H

Positive response

“P”	54H	32H	st0	st1	st2	Sti

Negative response

“N”	54H	32H	e1	e0

This command is used to inquire status of I2C card and return status by Sti.

Sti meanings:

Sti=30 H No I2C be activated
Sti=31 H Activated 24C01
Sti=32 H Activated 24C02
Sti=33 H Activated 24C04
Sti=34 H Activated 24C08
Sti=35 H Activated 24C16
Sti=36 H Activated 24C32
Sti=37 H Activated 24C64
Sti=38 H Activated 24C128
Sti=39 H Activated 24C256

3.5.4 I2C operation：

These functions are specified by a command data form like C-APDU which format is based on ISO/IEC 7816 T=0 standard.

In this case, CRT-591 recognizes the meaning of the command data, and execute the treatment related to the card by controlling hardware.

After the command was executed properly, CRT-591 returns a positive response with response data 9000H like from the IC card. When an error occurs during the communication with I2C, CRT-591 returns a positive response with status information in response data "sw1+sw2” which is base on ISO/IEC 7816-3 T=0

Sw1	Sw2	Description
90H	00H	Success
6FH	00H	Fail
6BH	00H	Address overflow
67H	00H	Operation length overflow

Write/Read I2C and Address scope is showed below:

Card_type	ab,cd
24C01	0000H ~ 007FH
24C02	0000H ~ 00FFH
24C04	0000H ~ 01FFH
24C08	0000H ~ 03FFH
24C16	0000H ~ 07FFH
24C32	0000H ~ 0FFFH
24C64	0000H ~ 1FFFH
24C128	0000H ~ 3FFFH
24C256	0000H ~ 7FFFH

3.5.4.1 Read data from I2C:

HOST Command

“C”	54H	33H	00H	B0H	abH	cdH	efH

Positive response

“P”	54H	33H	st0	st1	st2	Data

Negative response

“N”	54H	33H	e1	e0

Value:

ab: The upper address of head address which begins to read data

cd: The lower address of head address which begins to read data

ef: The number of bytes of data to read

CRT-591 read efH length and return to HOST according to address specified by abH, cdH.

The length of efH can not be surpass the length of I2C address up limit.

When the following command is transmitted, data can be read from the I2C memory card. （read data on 8bytes from the card）

Ex). “CU3”+00B0000008

3.5.4.2 Write data to I2C:

HOST command:

“C”	54H	34H	00H	D0H	abH	cdH	efH	ghH…

Positive response:

“P”	54H	34H	st0	st1	st2	Data

Negative response:

“N”	54H	34H	e1	e0

Value： ab: The upper address of head address which begins to write data

cd： The lower address of head address which begins to write data

ef: The number of bytes of data to write

gh: write data (efH byte)

CRT-591 write efH length and return to HOST according to address specified by abH, cdH.

The length of efH can not be surpass the length of I2C address up limit.

Ex). “CU3”+00B0000008+ write data(8 byte)

After command execute, return 9000H (operation success) or sw1,sw2(operation failure) result.

3.6 Contactless IC card Operation：

3.6.1 Activated contactless IC card：

HOST **Command**

“C”	60H	30H	Set1	Set2

(1) Mifare One Card Positive Response

“P”	60H	30H	st0	st1	st2	Rtype	ATQA	UID_len	UID_data	SAK

Mifare One Card Negative Response

“N”	60H	30H	e1	e0	Rtype	ATQA	UID_len	UID_data	SAK	

(2) 14443 Type A Card Positive Response

“P”	60H	30H	st0	st1	st2	Rtype	ATQA	UID_len	UID_data	SAK	ATS

14443 Type A Card Negative Response

“N”	60H	30H	e1	e0	Rtype	ATQA	UID_len	UID_data	SAK	ATS	

(3) 14443 Type B Card Positive Response

“P”	60H	30H	st0	st1	st2	Rtype	ATQB

14443 Type b Card B Negative Response

“N”	60H	30H	e1	e0	Rtype	ATQB

Activate RFID card

CRT-591-M support activated IEC/ISO14443 Type A and IEC/ISO 14443 Type B

The process is show as below:

1) Mifare one card:

1. Request A (REQ A) / Answer Request A (ATQ A).
2. Anticollision
3. Select (SEL) / Unique Identifier (UID) & Select Acknowledge(SAK)

When Mifare card successfully activate, CRT-591return:

ATQA (2 byte), UID_data (4—10 byte) and SAK(1 byte).

2) ISO/IEC 14443 Type A:

1. Request A(REQ A) / Answer Request A (ATQ A).
2. Anticollision
3. Select(SEL) / Unique Identifier(UID) & Select Acknowledge(SAK)
4. Request for answer to select (RATS) / Answer to Select(ATS)
5. Protcol and parameter selection request(PPSR) / PPS start(PPSS

When ISO/IEC 14443 Type A card successfully activated, CRT-591 return:

Mifare card return value increase (ATS(1-254 byte) and protocol parameter (1 byte))

3) ISO/IEC 14443 Type B:

1. Request B(REQ B) / Answer Request B (ATQ B).
2. Attribute(A TTRIB) / Answer to ATTRIB

When ISO/IEC 14443 Type B card successfully activated, CRT-591 return ATQB 12 byte(including following information):

50H, PUPI(4 byte) , App.data(4 byte), Protoclol info(3 byte)

Notes:

Set1,Set2 set sequence of operation for different type of protocol

Activate sequence： Type A protocol (first sequence), Type B protocol (second sequence)

Ex2：Set1=‘B’，Set2 =‘A’

Activate sequence： Type B protocol (first sequence), Type A protocol (second sequence)

Ex3：Set1=‘A’，Set2 =‘0’

Activate sequence： Type A protocol (first sequence), Type B protocol (Deactivated)

Ex4：Set1=‘B’，Set2 =‘0’，

Activate sequence： Type B protocol (first sequence), Type A protocol (Deactivated)

Rtype: Protocol

= 41H (‘A’) In line with ISO/IEC 14443 Type A protocol

= 42H (‘B’) In line with ISO/IEC 14443 Type B protocol

= 4DH (‘M’) In line with Philps Mifare one card protocol

Rtype=4DH(‘M’) : When Rtype=4DH(‘M’)

ATQA= 0044H Mifare Ultralight Card

ATQA= 0004H Mifare S50 1K Card

ATQA= 0002H Mifare S70 4K Card

Mifare one, ISO/IEC 14443 Type A return UID (The length of UID_data)

UID_len=4 The length of UID_data is 4 byte

UID_len=7 The length of UID_data is 7 byte

UID_len=10 The length of UID_data is10 byte

3.6.2 Deactivate RFID card：

HOST Command

“C”	60H	31H

Positive response

“P”	60H	31H	st0	st1	st2

Negative response

“N”	60H	31H	e1	e0

Deactivate RFID card and Output signal to antenna is closed。

3.6.3 Inquire status of RFID card：

HOST Command

“C”	60H	32H

Positive response

“P”	60H	32H	st0	st1	st2	sti	stj

Negative response

“N”	60H	32H	e1	e0

Inquire status of RFID sti,stj：

sti	stj	Specification
‘0’	‘0’	Deactivated RF
‘1’	‘0’	Mifare one S50 card
	‘1’	Mifare one S70 card
	‘2’	Mifare one UL card
‘2’	‘0’	Type A CPU card
‘3’	‘0’	Type B CPU card

3.6.4 Mifare 1 card control

These functions are specified by a command data form like C-APDU which format is based on T=0 standard. In this case, CRT-591 recognizes the meaning of the command data, and execute the treatment related to the card by controlling hardware. After the command was executed properly, CRT-591 returns a positive response with response data 9000H like from the IC card. When an error occurs during the communication with Mifare 1 card CRT-591 returns a positive response with status information in response data "sw1+sw2" which is base on ISO/IEC 7816-3.

Sw1	Sw2	Specification
90H	00H	Success
6FH	00H	Fail
6BH	00H	Address overflow
67H	00H	Operation length overflow

3.6.4.1 Key verification:

HOST Command

"C"	60H	33H	00H	20H	ks	sn	lc	pdata

Positive response

"P"	60H	33H	st0	st1	st2	rdata

Negative response

"N"	60H	33H	e1	e0

Download key to CRT-591 and verify the key directly

ks(1byte): key select (Key A=00H，Key B=01H)

sn(1byte): sector number (S50 card sn=00H-0FH, S70 card sn=00H-27H)

lc(1byte): password length lc=06H

pdata(6 byte): password data

rdata(2 byte): return data

return data(positive response with data 9000H, and negtive response with " sw1+sw2").

3.6.4.2 Verify key from EEPROM:

HOST Command

"C"	60H	33H	00H	21H	ks	sn

Positive response

"P"	60H	33H	st0	st1	st2	rdata

Negative response

“N”	60H	33H	e1	e0

Read key from EEPROM of RF module and verify the sector key

Download key via command mentioned in 9.10.4.4

EEPROM can preserve 32 groups of key data

ks(1byte): key select (Key A=00H，Key B=01H)

sn(1byte): sector number (sn=00H-0FH)

rdata(2 byte): return data positive response with 9000H

3.6.4.3 Modify sector key (KEY A):

HOST Command

“C”	60H	33H	00H	D5H	00H	sn	lc	pdata

Positive response

“P”	60H	33H	st0	st1	st2	rdata

Negative response

“N”	60H	33H	e1	e0

Modify sector key (key A)

This command only can modify KEY A, an d modify KEY B as“0xFF, 0xFF, 0xFF,0xFF,0xFF,0xFF in the mean timemodify control words as“0xFF, 0x07, 0x80, 0x69” (ex-work default)

Use block command to modify Key A, Key B control word

sn(1byte): sector number (S50 card sn=00H-0FH, S70 card sn=00H-27H)

lc(1byte): Password length lc= 06H

pdata ： password data 6 byte.

rdata(2 byte): positive response with data 9000H, and negtive response with “ sw1+sw2

3.6.4.4 Download password to EEPROM:

HOST Command

“C”	60H	33H	00H	D0H	ks	sn	lc	pdata

Positive response

“P”	60H	33H	st0	st1	st2	rdata

Negative response

“N”	60H	33H	e1	e0

Read key from EEPROM of RF module and verify the sector key

EEPROM can preserve 32 groups of key data

ks(1byte): key select (Key A=00H， Key B=01H)

sn (1byte): sector number (sn=00H-0FH)

lc(1byte): password length lc=06H

pdata(6 byte): password data

rdata(2 byte): return data

positive response sw1+sw2=9000H.
negtive response sw1+sw2=6F00H

3.6.4.5 Read sector data:

HOST Command

"C"	60H	33H	00H	B0H	sn	bn	le

Positive response

"P"	60H	33H	st0	st1	st2	rdata

Negative response

"N"	60H	33H	e1	e0

Read block and sequence blocks from RF card

sn(1 byte): sector number
bn(1 byte): block number
le(1 byte): block number (le=01H read one block, le=03H read three blocks)
rdata(2 byte): return data positive response with data 9000H, and negtive response with " sw1+sw2

Notes：

1. Ultralight Card only have one block in one sector， every block have 4 byte data. S50, S70 have 16 byte data in one block.
2. Ultralight Card, Mifare 1k (S50), Mifare 1k (S70) card range of capacity is shown as below
Ultralight Card: sn=00H-0FH, bn=00H, le=01H-0FH
Mifare 1k(S50): sn=00H-0FH, bn=00H-03H, le=01H-04H
Mifare 1k(S70): sn=00H-20H, bn=00H-03H, le=01H-04H
sn=21H-27H, bn=00H-0FH, le=01H-10H (S70 card last 8 sector have 16 blocks)

3.6.4.6 Write sector data:

HOST Command

"C"	60H	33H	00H	D1H	sn	bn	lc	wdata

Positive response

"P"	60H	33H	st0	st1	st2	rdata

Negative response

"N"	60H	33H	e1	e0

Read block and sequence blocks from RF card

sn(1 byte): sector number
bn(1 byte): block number
le(1 byte): block number
wdata: block to write (n byte)
rdata(2 byte): return data positive response with data 9000H, and negtive response with " sw1+sw2

Notes：

1. Ultralight Card only have one block in one sector，every block have 4 byte data. S50,S70 have16 byte data in one block
2. Ultralight Card,Mifare 1k(S50), Mifare 1k (S70) card card range of capacity is shown as below:

Ultralight Card: sn=00H-0FH, bn=00H-03H, lc=01H-03H
Mifare 1k(S50): sn=00H-0FH, bn=00H-03H, lc=01H-03H
Mifare 1k(S70): sn=00H-20H, bn=00H-03H, lc=01H-03H
sn=21H-27H, bn=00H-0FH, lc=01H-0FH
(S70 card last 8 sector have 16 blocks)

3. S50,S70 card last block of each sector is control sector to preserve Key A, read/write control words, Key B.

Cautions: Do note write last block and CRT-591M also will prohibid to write last block

3.6.4.7 Initialization:

HOST Command

“C”	60H	33H	00H	D2H	sn	bn	lc	wdata

Positive response

“P”	60H	33H	st0	st1	st2	rdata

Negative response

“N”	60H	33H	e1	e0

Initialization operation to RF card

sn(1 byte): sector number
bn(1 byte): block number
lc(1byte): lc=04H Initialization data length 1c=04H
wdata: Initialization data (4 byte)
rdata(2 byte): return data (positive response with data 9000H,and negative response with sw1+sw2(2 byte)

Notes：Mifare 1k(S50), Mifare 1k (S70) card operation sector (Sector can not be out of range and last block can not be operated)

Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k(S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,
(S70 card last 8 sector have 16 blocks)

3.6.4.8 Read value

HOST Command

“C”	60H	33H	00H	B1H	sn	bn

Positive response

“P”	60H	33H	st0	st1	st2	rdata

Negative response

“N”	60H	33H	e1	e0

Read value operations to RF card

sn(1 byte): sector number
bn(1 byte): block number

rdata: return data (positive response with data (4 byte)+9000H, and negtive response with " sw1+sw2" (2 byte))

Notes：Mifare 1k(S50), Mifare 1k (S70) card operation sector

(Sector can not be out of range and last block can't be operated)

Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k(S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,

(S70 card last 8 sector have 16 blocks)

3.6.4.9 Increment:

HOST Command

"C"	60H	33H	00H	D3H	sn	bn	lc	wdata

Positive response

"P"	60H	33H	st0	st1	st2	rdata

Negative response

"N"	60H	33H	e1	e0

Increment operation to RF card
sn(1 byte): sector number
bn(1 byte): block number
lc(1byte): increment length lc=04H
wdata: increment data(4 byte)
rdata: return data

(positive response with data 9000H, and negtive response with " sw1+sw2")

Notes：Mifare 1k(S50), Mifare 1k (S70) card operation sector

(Sector can not be out of range and last block can not be operated and the last block of each sector cannot be value operated on)

Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k(S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,

(S70 card last 8 sector have 16 blocks)

3.6.4.10 Decrement:

HOST Command

"C"	60H	33H	00H	D4H	sn	bn	lc	wdata

Positive response

"P"	60H	33H	st0	st1	st2	rdata

Negative response

"N"	60H	33H	e1	e0

Decrement operation to RF sector
sn(1 byte): sector number
bn(1 byte): block number
lc(1byte): Decrement length lc=04H

wdata: Decrement data(4 byte)

rdata: return data

(positive response with data 9000H, and negtive response with " sw1+sw2" (2 byte)

Notes：Mifare 1k(S50), Mifare 1k (S70) card operation sector

(Sector can not be out of range and last block can not be operated)

Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,

Mifare 1k(S70): sn=00H-20H, bn=00H-03H,

sn=20H-27H, bn=00H-0EH, (S70 card last 8 sector have 16 blocks)

3.6.4.11 ULC Card Authentication (Customized Model)

HOST command:

"C"	60H	33H	00H	30H	secret key

Positive return:

"P"	60H	33H	st0	st1	st2	rdata

Negative return:

"N"	60H	33H	e1	e0

Perform authentication operations on the ULC card..

secret key (16 byte): Authentication Key

arrangement:

Ⅰ. The 16-byte Key is divided into two groups,with each group consisting of 8 bytes.The Key for pages 44 and 45 in the card storage area forms one group,and the Key for pages 46 and 47 forms the other group.

Ⅱ.Each group is arranged with the higher address first and the lower address last.

The initial key for a typical ULC card is:

0x49,0x45,0x4D,0x4B,0x41,0x45,0x52,0x42,0x21,0x4E,0x41,0x43,0x55,0x4F,0x59,0x46.

rdata: return data The result of the operation.A successful operation returns 9000H..

In case of operation failure,only sw1+sw2(2 bytes)are returned..

Note:This command is only supported by customized models.

3.6.4.12 ULC Card Permission Settings (Customized Models)

HOST command:

"C"	60H	33H	00H	31H	permissions	StarPage

Positive return:

"P"	60H	33H	st0	st1	st2	rdata

Negative return:

"N"	60H	33H	e1	e0

Perform operations on the access permissions of ULC card sectors..

permissions (1 byte): 0xF1

Permission	permissions	StarPage
Read-only	0xF1	0x00
Read/Write	0xF2	0x00

No Permission	0xF1	0x30

StarPage (1 byte)When reading and writing:The permission starts from page n. n<=0x30
No permission:Fixed as 0x30.

rdata: return data The result of the operation.A successful operation returns 9000H..
In case of operation failure,only the following is returned: sw1+sw2(2 byte).

Note:This command is only supported by customized models.

3.6.4.13 ULC Card Key Setting (Customized Models)

HOST command:

"C"	60H	33H	00H	32H	secret key

Positive return:

"P"	60H	33H	st0	st1	st2	rdata

Negative return:

"N"	60H	33H	e1	e0

Modify the ULC Card Sector Key.

Secret Key(16 bytes):Key

Arrangement Order:

I.The 16-byte Key is divided into two groups,with each group consisting of 8 bytes.The Key for pages 44 and 45 in the card storage area forms one group,and the Key for pages 46 and 47 forms the other group.

II.Each group is arranged with the higher address first and the lower address last.

rdata:Return data

The result of the operation.A successful operation returns 9000H.

In case of operation failure,only sw1+sw2(2 bytes)are returned.

Note:This command is only supported by customized models.

3.6.4.14 Ultralight AES operation (Customized model support)

3.6.4.14.1 Ultralight AES Authentication

Host command:

"C"	60H	33H	00H	50H	30H	KeyType	secret key(16 Byte)

Positive return:

"P"	60H	33H	st0	st1	st2	rdata

Negative return:

"N"	60H	33H	e1	e0

Perform authentication operations on the card.

KeyType:Key type
31H:DataProtKey
32H:UIDRetrKey
Secret Key(16 bytes):Authentication Key
Arrangement Order:

I.The 16-byte Key is arranged with the higher address first and the lower address last.

Page address	Byte Number			
Hex	Byte 0	Byte 1	Byte 2	Byte 3
30h	K0	K1	K2	K3
31h	K4	K5	K6	K7
32h	K8	K9	K10	K11
33h	K12	K13	K14	K15

If the values in the above figure are K0=0x01,K1=0x02,...,K15=0x10,then the authentication command should be sent.:

43h 60h 33h 00h 50h 30h 31h 10h 0Fh 0Eh 0Dh 0Ch 0Bh 0Ah 09h 08h 07h 06h 05h 04h 03h 02h 01h

rdata: return data The result of the operation.A successful operation returns 9000H.
In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.2 Obtain card information.

Host command:

“C”	60H	33H	00H	50H	31H

Positive return:

“P”	60H	33H	st0	st1	st2	Cardinfo(8byte)	rdata

Negative return:

“N”	60H	33H	e1	e0

Cardinfo (8byte):

Byte no.	Description	MIFARE Ultralight AES	Interpretation
0	fixed header	00h	
1	vendor ID	04h	NXP Semiconductors
2	product type	03h	MIFARE Ultralight
3	product subtype	01h / 02h	17 pF / 50 pF
4	major product version	04h	AES
5	minor product version	00h	V0
6	storage size	0Fh	144-byte
7	protocol type	03h	ISO/IEC 14443-3 compliant

rdata: return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.3 Read the counter value

Host command:

“C”	60H	33H	00H	50H	33H	Addr

Positive return:

“P”	60H	33H	st0	st1	st2	Value(3byte)	rdata

Negative return:

“N”	60H	33H	e1	e0

Addr: counter number from 00h to 02h
Rdata: Return a 3-byte count value

rdata: return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.4 Increase the count value

Host command:

“C”	60H	33H	00H	50H	34H	Addr	IncrValue(3Btye)

Positive return:

“P”	60H	33H	st0	st1	st2	rdata

Negative return:

“N”	60H	33H	e1	e0

Addr: counter number from 00h to 02h
IncrValue: Range of increased values 000000H~FFFFFFH

rdata:return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.5 Read ECC signature

Host command:

“C”	60H	33H	00H	50H	35H

Positive return:

“P”	60H	33H	st0	st1	st2	ECC signature	rdata

Negative return:

“N”	60H	33H	e1	e0

ECC signature: return 48byte

rdata:return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.6 Write ECC signature

Host command:

“C”	60H	33H	00H	50H	36H	Addr	ECC Sig(4byte)

Positive return:

“P”	60H	33H	st0	st1	st2	rdata

Negative return:

“N”	60H	33H	e1	e0

Addr:BlockAddress(1Byte) range:00h-0Bh
ECC Sig: signature bytes to be written(4Byte)

rdata:return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.7 Lock ECC signature

Host command:

“C”	60H	33H	00H	50H	37H	locking action

Positive return:

“P”	60H	33H	st0	st1	st2	rdata

Negative return:

“N”	60H	33H	e1	e0

locking action：1byte

=00h unlock
=01h lock
=02h permanent lock

rdata: return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.8 Key modification

Host command:

“C”	60H	33H	00H	50H	38H	KeyType	KeyDate

Positive return:

“P”	60H	33H	st0	st1	st2	rdata

Negative return:

“N”	60H	33H	e1	e0

KeyType:1 Byte
=31h DataProtKey
=32h UIDRetrKey
KeyDate:16Byte modified key
arrangement:
The 16-byte key is arranged with the higher address at the front and the lower address at the back.

Page address	Byte Number			
Hex	Byte 0	Byte 1	Byte 2	Byte 3
30h	K0	K1	K2	K3
31h	K4	K5	K6	K7
32h	K8	K9	K10	K11
33h	K12	K13	K14	K15

If the values in the above figure are K0=0x01,K1=0x02,...,K15=0x10,then the command to modify the key should be sent.:
43h 60h 33h 00h 50h 38h 31h 10h 0Fh 0Eh 0Dh 0Ch 0Bh 0Ah 09h 08h 07h 06h 05h 04h 03h 02h 01h

rdata: return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned.

3.6.4.14.9 Configure sector access permissions

Host command:

“C”	60H	33H	00H	50H	39H	Pm	Page Address

Positive return:

“P”	60H	33H	st0	st1	st2	rdata

Negative return:

“N”	60H	33H	e1	e0

Pm:1Byte

=30h Revoke permissions.The command to revoke permissions does not require sending PageAddress data.

=31h Configure permissions

PageAddre (1Byte): The starting sector that requires permission,with a range of 00h to 3Ch.This command configures the range of permissions.

The default termination address within the system is 3Ch,and the starting address is the specified PageAddress.

When PageAddress is set to 3Ch,it is equivalent to revoking permission.

rdata: return data The result of the operation.A successful operation returns 9000H. In case of operation failure,only sw1+sw2(2 bytes)are returned..

3.6.5 Type A RF card communication:

3.6.5.1 APDU Operate

HOST Command

“C”	60H	34H	C-APDU

Positive response

“P”	60H	34H	st0	st1	st2	R-APDU

Negative response

“N”	60H	34H	e1	e0

This exchanges data between RF card by protocol RF Type A T=CL according to ISO/IEC 14443-4

Notes: The max. length of C-APDU is 261 byte, the max. length of R-APDU is 258 byte.

3.6.6 Type B RFcard communication:

3.6.6.1 APDU Operate

HOST Command

“C”	60H	35H	C-APDU

Positive response

“P”	60H	35H	st0	st1	st2	R-APDU

Negative response

“N”	60H	35H	e1	e0

This exchanges data between RF card by protocol RF Type B T=CL according to ISO/IEC 14443-4

Notes: The max. length of C-APDU is 261 byte, the max. length of R-APDU is 258 byte.

3.6.7 Operations on cards using the 15693 protocol

Supported memory card size:Number of blocks:<=256,Block size:4 bytes

3.6.7.1 Read block operation

Host command:

“C”	60H	36H	30H	Sn	Le

Positive return:

“P”	60H	36H	st0	st1	BlockData	rdata

Negative return:

“N”	60H	36H	e1	e0

Read a block of data or multiple consecutive blocks of data from the specified block

Sn(1Byte) Start Block Number sn:0-255

Le(1Byte) Continuous block read length(Le=01H to read one block,Le=02H to read two blocks,Le=0 to continuously read 256 blocks)

Blockdata:Returned block data, length=4(Byte)*Le

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.2 Write block operation

Host command:

“C”	60H	36H	31H	Sn	Le	wdara

Positive return:

“P”	60H	36H	st0	st1	rdata

Negative return:

“N”	60H	36H	e1	e0

Write a single block of data or multiple consecutive blocks of data starting from the specified block.

Sn(1 Byte):Starting block number,where sn ranges from 0 to 255.

Le(1 Byte):Length of consecutive blocks to write(Le=01H to write one block,Le=02H to write two blocks,Le=0 to continuously write 256 blocks).

wdata:Data to be written to the block(s),with the data length equal to 4(Bytes)×Le.

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.3 Get system information

Host command:

“C”	60H	36H	32H

Positive return:

“P”	60H	36H	st0	st1	Information sign	UID	DSFID	AFI	Total blocks	Size of each block	IC benchmark	rdata
					1Byte	8Byte	1Byte	1Byte	1Byte	1Byte	1Byte	

Negative return:

“N”	60H	36H	e1	e0

Information Flag Definition (The returned card information is defined according to the following flags for return):

bit	Sign	Value	Description
Bit0	DSFID	0	The DSFID is not supported,and the DSFID field does not exist.
		1	The DSFID is supported,and the DSFID field exists.
Bit1	AFI	0	The AFI is not supported,and the AFI field does not exist.
		1	The AFI is supported,and the AFI field exists.
Bit2	IC card memory capacity	0	Memory information is not supported, and the memory information field does not exist.

		1	Supports memory information, and the memory information field exists.
Bit3	IC benchmark	0	IC benchmark is not supported, IC benchmark field does not exist
		1	Supports IC benchmark, IC benchmark field exists
Bit4	reservation	-	---
Bit5	reservation	-	---
Bit6	reservation	-	---
Bit7	reservation	-	---

UID:Unique identifier for the IC card.If present,it occupies 8 bytes.

DSFID:Data Storage Format Identifier.If present,it occupies 1 byte.

AFI:Application Family Identifier.If present,it occupies 1 byte.

Number of Blocks:The quantity of blocks.If present,it occupies 1 byte.

Block Size:The maximum capacity of each block.This device only supports reading and writing to IC cards with 4-byte blocks.If present,it occupies 1 byte.

IC Reference:Defined by the IC card manufacturer.If present,it occupies 1 byte.

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.4 Write AFI operation

Host command:

“C”	60H	36H	33H	31H	AFI(1Byte)

Positive return:

“P”	60H	36H	st0	st1	rdata

Negative return:

“N”	60H	36H	e1	e0

AFI:The AFI value to be written(1 Byte)

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.5 Lock AFI operation

Host command:

“C”	60H	36H	33H	32H

Positive return:

“P”	60H	36H	st0	st1	rdata

Negative return:

“N”	60H	36H	e1	e0

Note: This operation is irreversible.

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.6 Write DSFID operation

Host command:

“C”	60H	36H	34H	31H	DSFID(1Byte)

Positive return:

“P”	60H	36H	st0	st1	rdata

Negative return:

“N”	60H	36H	e1	e0

DSFID: The DSFID value to be written(1 Byte)

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.7 Lock DSFID operation

Host command:

“C”	60H	36H	34H	32H

Positive return:

“P”	60H	36H	st0	st1	rdata

Negative return:

“N”	60H	36H	e1	e0

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

Note: This operation is irreversible.

3.6.7.8 Obtain the block security status

Host command:

“C”	60H	36H	35H	Sn	Le

Positive return:

“P”	60H	36H	st0	st1	BlockStatus	rdata

Negative return:

“N”	60H	36H	e1	e0

Read the security status of a specified block or multiple consecutive blocks

BlockStatus:0H:Unlocked, 1H: Locked state.

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

3.6.7.9 Locking Block

Host command:

“C”	60H	36H	36H	Sn

Positive return:

“P”	60H	36H	st0	st1	BlockStatus	rdata

Negative return:

“N”	60H	36H	e1	e0

Lock the specified block

Sn(1 Byte):specified block number

Rdata(2Byte) return dataA successful operation returns the structure 9000H,while a failed operation returns sw1+sw2

Note: This operation is irreversible.

3.6.8 Read Serial number:

HOST Command

“C”	A2H	30H

Positive response

“P”	A2H	30H	st0	st1	st2	len	ICRW_SN

Negative response

“N”	A2H	30H	e1	e0

Len: read length of CRT-591serial number (0byte-18byte)

ICRW_SN: machine serial number

3.6.9 Read information of machine configuration

HOST Command

“C”	A3H	30H

Positive response

“P”	A3H	30H	st0	st1	st2	ICRW_Config

Negative response

“N”	A3H	30H	e1	e0

3.6.10 Error-card Bin Counter Control:

3.6.10.1 Read error-card bin counter

HOST Command

“C”	A5H	30H

Positive response

“P”	A5H	30H	st0	st1	st2	Count(3 byte)

Negative response

“N”	A5H	30H	e1	e0

After reset error-card bin counter. Capture on card, counter one plus

Count= “000” ~ “999”

Counter overflow will return machine status (e1,e0=“50”)

3.6.10.2 Set initial value of error-card bin

HOST Command

“C”	A5H	31H	Count(3 byte)

Positive response

“P”	A5H	31H	st0	st1	st2

Negative response

“N”	A5H	31H	e1	e0

Set initial value of error-card bin.

Count= “000” ~ “999”

Count value range (0-999)

3.6.11 Reset PC/SC reader (only available for CRT-591-MTX1)

HS Command

“C”	A6H	30H

Positive response

“P”	A6H	30H	st0	st1	st2

The command is to reset PC/SC reader。

3.7 Get Firmware Version

Host command:

“C”	DCH	31H

Positive return:

“P”	DCH	31H	st0	st1	st2	Version

Negative return:

“N”	DCH	31H	e1	e0

Get the firmware version, such as:C591_MR01_A_150123